ARTICLE

Autism and Considerations in Recreation and Physical Activity Settings

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Introduction

The Centers for Disease Control and Prevention reports that 1.5 million people who have some form of autism live in the United States. Also, 1 in every 166 children is diagnosed with autism (CDC, 2007). Autism is the third most common developmental disability in the United States. The occurrence of autism in males is significantly higher than in females, currently at a 4:1 ratio, and for children with severe to profound intellectual disability, the ratio is 2:1 (Fombonne, 2002). In the most current edition of the Diagnostic and Statistical Manual-DSM IV TR (American Psychiatric Association, 2000), autism is the most common of the five Pervasive Developmental Disorders (PDD). PDD is a category of neurological disorders with severe and pervasive impairments affecting the development of social interactions and communication skills. The other four are Rett's Disorder, Child Disintegrative Disorder, Asperger's Disorder and Pervasive Developmental Disorder Not Otherwise Specified (PDD-NOS).

Autism is a spectrum of conditions of neurobiological origin unrelated to parental nurturance with no known cause. It appears to be genetically linked and is a lifelong disorder (Bailey et al., 1995; International Molecular Genetic Study of Autism Consortium, 2001). Development in three main areas is impaired: verbal and non-verbal communication skills, social skills with limited or no initiation or reciprocal interaction, and restricted range of play and interests, including repetitive behaviors. Impairments may range from mild to severe, and each area of development may be affected to a different degree in the same child, where one area of development may be mildly impaired, while another area may be severely impaired. No two children are alike (American Psychiatric Association, 2000).

Presently there are no medical or biological tests to diagnose autism. The diagnosis is based on history, diagnostic tools and observations. A diagnosis of autism requires a comprehensive evaluation by a multidisciplinary team that usually includes a developmental pediatrician, psychologist, psychiatrist, physical, occupational, speech, behavioral and developmental therapist, a social worker, and others.

Autism is generally evident by age 3. In 1990, the average age of diagnosis was 4-6 years and in 2000, the age was 30 months (Neurology, 2000). An early diagnosis is important for the attainment of services and improved outcomes. Early intervention of services has shown to be effective in improving the development of children with autism and the prevention of secondary conditions (Filipek, Accardo, et al., 1999). Parent involvement is crucial.

Associated conditions can include, but are not limited to:

- Cognitive impairment
- Seizures

Secondary conditions can include:

- Anxiety
- Depression
- Obesity

The three characteristics of autism — 1) communication, 2) social interactions, and 3) restrictive play and interest in addition to modification of the environment that is conducive to learning the skill or participating in the activity — are key areas to address and to be aware of when individuals with autism are involved in physical activity or recreation programs.

Playing in the Park: A Child with Autism Plays in the Park – Video 1 +

Playing in the Park: A Child with Autism Plays in the Park – Video 2 +

Part 2: Sensory Considerations

Individuals with autism may have hypoactive (under-responsive) or hyperactive (over-reactive) responses to any of the senses, including sounds, touch, taste, and smell. Sensory integration, the process in which the brain organizes and interprets sensory information, is often impaired in individuals with autism. Being aware of sensory processing difficulties that an individual may have and incorporating individualized sensory activities into his or her daily activities can benefit the individual with autism.

Beyond the basic five senses — vision, hearing, touch, taste, and smell — there are two other sensory processing systems, the proprioceptive and vestibular systems, that may be affected. The proprioceptive system helps the brain know where the body is in space. It is the deep pressure felt when we carry heavy bags, push big pieces of furniture, or go swimming. Proprioceptive stimulation can be the most calming and sought-after stimulation. Individuals whose proprioceptive system is affected may complain of being tired, move slowly and have difficulty participating in sports. The vestibular system helps us keep our balance, coordinate movements of the eyes, head, body movement and gravity and affects visual perception, auditory language perception, balance, muscle tone, and motor planning. Individuals whose vestibular system is affected may have a difficult time filtering out non-important information, maintaining eye contact, and variable issues.

Playing in the Park: A Child with Autism Plays in the Park – Video 3 +

Part 3: Behavioral Considerations

Behavior is often a method of communication. This is especially true for people with autism who may not be able to communicate their needs, feelings, or sensory concerns in typical ways. Their behaviors may be a response to an unmet need or a delayed ability or skill (cognitive or language delay or sensory issue) and serve as a function to get their needs met. In order to decrease the likelihood of challenging behaviors, it is important to plan the activity or exercise according to the individual's developmental level and to meet their sensory needs. For example, plan activities or exercise that capitalize on an individual's need to exhibit excessive movement by including activities that provide the movement he or she needs to stimulate the body and brain to stay alert and active.

Be aware of the potential for restricted, repetitive patterns of behaviors which include, but are not limited to, spinning, rocking, and hand-flapping. Working with, instead of against, some of these behaviors will benefit all involved. For example, if a child throws objects, involving that child in a game that requires throwing, such as bean-bag toss or basketball, will redirect the behavior to functional use. This may require pre-planning for potential issues that may arise.

Developing a plan for behavior management will provide a successful experience that meets the needs of the individual. Visual rules, instructions, and other supports may need to be developed to make expectations clear and concrete. Structured and predictable routines and environments are essential for a child with autism to be successful in participating in activities. There will also be varying degrees of severity and rigid behavior. Strategies for behavior management can include:

- Make use of need for sameness and structure when setting up a consistent exercise plan or activity routine.
- Remove all extra stimuli to reduce distractions and sensory overload and/or set up boundaries to minimize distractions.
- Provide predictable, structured transitions from one activity or event to another.
- Utilize visual directions, such as picture exchange systems and visual picture schedules, and limit verbal directions to a few words or short sentences.
- Be mindful of the length of activities because some children may need frequent changes due to limited attention span, while others, may not.
- Chose safe activities as individuals may not be aware of danger.

Playing in the Park: A Child with Autism Plays in the Park – Video 4 +

Playing in the Park: A Child with Autism Plays in the Park – Video 5 +

Part 4: Social Considerations

Individuals with autism typically socialize differently than individuals without autism. They may demonstrate difficulty in perceiving other people's needs, thoughts, and feelings, or they may assume the literal meaning of words rather than deciphering subtle social cues and gestures. Maintaining a conversation can be a challenge for individuals with autism as their interest in other people may be reduced by a preference to be alone. Maintaining eye contact can also be difficult due to sensory stimulation, but this is not always the case. Social communication, both verbal and non-verbal, is impaired in individuals with autism; therefore, group dynamics and cooperation may be difficult as social skills, and social cues are needed for appropriate interactions. Since social interaction skill development is often a significant concern for individuals with autism, it is very important to teach social skills during physical and recreational activities, which typically include social components. Social stories — stories written for an individual with autism to learn how to handle challenging or new social situations — have shown positive results (Gray, 1996). Because social skills are developed throughout a person's life and are necessary for social interaction, developing friendships and social relationships, incorporating social skills teaching and providing opportunities to practice these skills is necessary for lifelong skill attainment.

Playing in the Park: A Child with Autism Plays in the Park – Video 6 +

Playing in the Park: A Child with Autism Plays in the Park – Video 7 +

Part 5: Cognition Considerations

Persons with autism are often visual learners and greatly benefit from visual instruction using visual cues, individual picture schedules, and work systems. Use of video or video modeling to learn skills can also be a source of visual instruction. Treatment and Education of Autistic and Related Communication Handicapped Children (TEACCH) approach has had significant success with individuals who have autism (Schopler, et. al., 1995). TEACCH is a structured teaching approach that utilizes visual structure. Auditory instruction is how most people learn. This is often difficult for people with autism.

IQ scores for individuals with autism range from high intelligence to severe and profound intellectual disability. Short attention span, difficulty attending to multiple cues, focusing too much on details, and information processing impairments with difficulty integrating, sequencing, organizing, and transferring of information from one situation to another can be present.

Part 6: Speech, Hearing and Communication Considerations

A child with autism may have difficulty expressing needs, initiating and maintaining conversations, and have repetitive use of language (echolalia). Keep in mind that speech is the actual production of sounds that produce words, language is the underlying meaning behind those words, and communication is any modality that is used to give and receive a message. Thus, communication can be verbal and nonverbal. Hearing is frequently one of the parent's initial concerns, although tests often show that it falls within normal limits. However, despite adequate hearing, individuals' responses to auditory information are frequently poor.

Playing in the Park: A Child with Autism Plays in the Park – Video 8 +

Part 7: Neurological Considerations

Seizures, which are an abnormality in the electric activity in the brain, occur in one of every four children with autism and will range in type and severity (Volkmarr, 2000). Persons who have seizures can engage in most activities, yet it is important to know if their seizures are controlled through medications, which may cause side effects. Medications that control seizures may have an effect on an individual's physiological response to exercise and activity. In addition, certain activities are not recommended for individuals who have seizures, such as boxing, soccer and underwater swimming. Other activities, such as swimming, horseback riding, and contact sports, should be monitored to ensure the individual's safety. Research suggests that exercise which is performed on a regular basis has been shown to inhibit seizure activity (Durstine & Moore, 2003).

Part 8: Inclusion

Inclusion, or the integration of people with and without disabilities, is supported by legislation of the Americans with Disabilities Act (ADA) and the Individuals with Disabilities Education Act (IDEA). A key component of IDEA addresses the need for individuals to receive services in the least restrictive setting or environment. Physical activity and recreation programs should be naturally inclusive environments for individuals with autism. To ensure a safe and successful program, staff need to be knowledgeable on the considerations of autism, and sufficiently trained on working and interacting with individuals with autism.

Part 9: Exercise Guidelines

Exercise and physical fitness are important components of a healthy lifestyle. This is especially true for individuals with autism, as children with autism have been found to have low levels of physical fitness (Auxter, Pyfer & Huettig, 1997). Physical activity can be instrumental for a person with autism to assist with his or her sensory integration, coordination, muscle tone, and social skills development. Yilmaz, Yanardag, Birkan, and Bumin (2004) found that cardiorespiratory, flexibility, balance, agility, and strength increased and stereotypical behaviors decreased after swimming.

It is important to develop an exercise program at a level comfortable for the individual. Intensity and duration of the exercise should be increased gradually overtime to accommodate the physiological training effects such as improved cardiovascular endurance, strength or flexibility. Introduce simple activities and progress to complex ones. Include activities that are enjoyable, and utilize the individual's interests and strengths when designing a physical activity program. An exercise program needs to address the four components of fitness, which include cardiovascular endurance, strength, flexibility, and balance. Considerations for each area of exercise programming are stated below.

Cardiovascular Guidelines

Cardiovascular exercise benefits a person's heart, lungs, and circulatory system, which affect endurance. Cardiovascular exercise is important for individuals with autism not only for physical fitness and the resulting health benefits, but because studies have indicated that vigorous cardiovascular exercise decreases inappropriate behaviors and increases appropriate behaviors (Watters & Watters, 1980; Kern, Koegel & Dunlap, 1984). Studies on participation in cardiovascular activities demonstrated improvements in attention span and on-task behavior (Power, Thibadeau & Rose, 1992) and decreased self-stimulating behavior (Rosenthal-Malek & Mitchell, 1997).

Research studies by Kern, Koegel, & Dunlap (1984) which incorporated using jogging versus mild physical activity that utilized ball playing as antecedents to influence subsequent stereotypical behaviors of autism have shown that stereotypical behaviors decreased after 15 minutes of continuous jogging, yet ball playing had no effect on these behaviors. Fatigue does not appear to be a factor, yet physiological and behavioral mechanisms appear to be factors influencing the reduction in stereotypical behaviors. Cardiovascular activities can also be instrumental in decreasing anxiety and depression, which individuals with autism are at a greater risk of experiencing.

Strength Training Guidelines

Strength training improves muscular strength and endurance. Core strength is especially important for individuals with autism, because their trunk muscles are typically weak. Core muscles have an effect on other activities, such as balance and coordination and an important foundation that must be established before working on higher-level skills. When developing a strength-training program, incorporate activities that focus on repetition of movements in the same order and remember to prepare the individual for changes in the program. Since individuals with autism respond best to structured learning, it is important to develop a sequential and consistent strength-training program. Visual cuing and exercise diagrams can be an important component of proper exercise form and foster the ability to exercise independently.

Flexibility Guidelines

Flexibility exercises encourage the movement of joints through a full range of motion. Flexibility is an important component of an overall physical activity program and an area that is important to address for individuals with autism as they often have low muscle tone. For example, individuals who have low muscle tone may walk on their toes to stimulate their proprioceptive system, which can result in hypertonicity or high tone in their calf muscles. Incorporating flexibility exercises which stretch the legs and feet are helpful.

Balance Guidelines

Balance is the ability to stabilize the body. Incorporating balance and coordination activities into a physical activity program is important for individuals with autism because it utilizes the vestibular system. Balance activities, such as using a therapy ball or walking a line on the floor, can improve balance and coordination. Individuals with autism may have a poor sense of direction and may need visual cues, such as standing on a carpet square or a mark on the floor to help them know where to stand.

Playing in the Park: A Child with Autism Plays in the Park – Video 9 +

Playing in the Park: A Child with Autism Plays in the Park – Video 10 +

Playing in the Park: A Child with Autism Plays in the Park – Video 11 +

Part 9: Physical Activity Schedule or Daily Routine

Every individual with autism is different; however, some physical activities, including activities that are sensory in nature, are important to incorporate daily. These activities can address physical recreation, sensory, and social needs. Each exercise plan should be tailored to the needs and likes of the individual. For example, one individual may need heavier and active routines incorporated into the mornings and more calming, soothing activities in the afternoons, while another may need calming activities throughout the day.

A model physical activity or recreation plan for individual with autism would include activities and exercises that:

- Are of interest and fun to the individual
- Work on core muscle strength
- Use the large muscle groups and provide continuous in motion, such as climbing, jumping, and running for children
- Include aquatics activities when appropriate, as it provides sensory motor exercise
- Build on the individual's strengths, such as martial arts, which offer visual cues with the instructor modeling movement or forms for students, or swimming, which can offer deep pressure for individuals who may need this type of sensory stimulation
- Promote positive experiences which an individual will want to repeat
- Develop lifelong leisure and physical activity skills
- Are well organized because unstructured time and waiting are usually difficult

Physical activity or recreation suggestions that can be incorporated into the daily routine:

- Walking, jogging, bike riding, or other physical activities in the morning, when possible, allows for a good start to the day.
- Allow children during their school schedule to move around the classroom, step out in the hall for a drink of water, or to deliver a message to a nearby classroom to incorporate some simple movement throughout the day.
- Jumping, pushing, swinging, and stretching can be calming and assist with re-focusing during the day.
- Yoga moves can be taught and used as calming and relaxing techniques and can be incorporated as soothing activities before or after transition times when appropriate.
- Carrying groceries, heavy bags, or heavy pushing (such as vacuuming) can be good afternoon activities.
- Swimming or participation in sport activities or modified sports can be excellent opportunities for incorporating physical and social activities.

Playing in the Park: A Child with Autism Plays in the Park – Video 12 +

Co-authors, Sheila Swann-Guerrero, CTRS, National Center on Health, Physical Activity, and Disability Information Specialist, and Martha Rubio-Cajigas, DT, Advocate Illinois Masonic Pediatric Developmental Center would like to thank the family featured in the autism factsheet video clips for their participation in the video filming.

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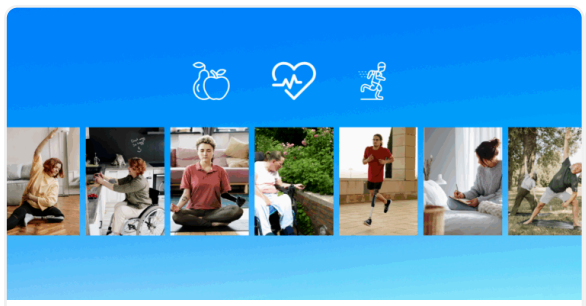

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